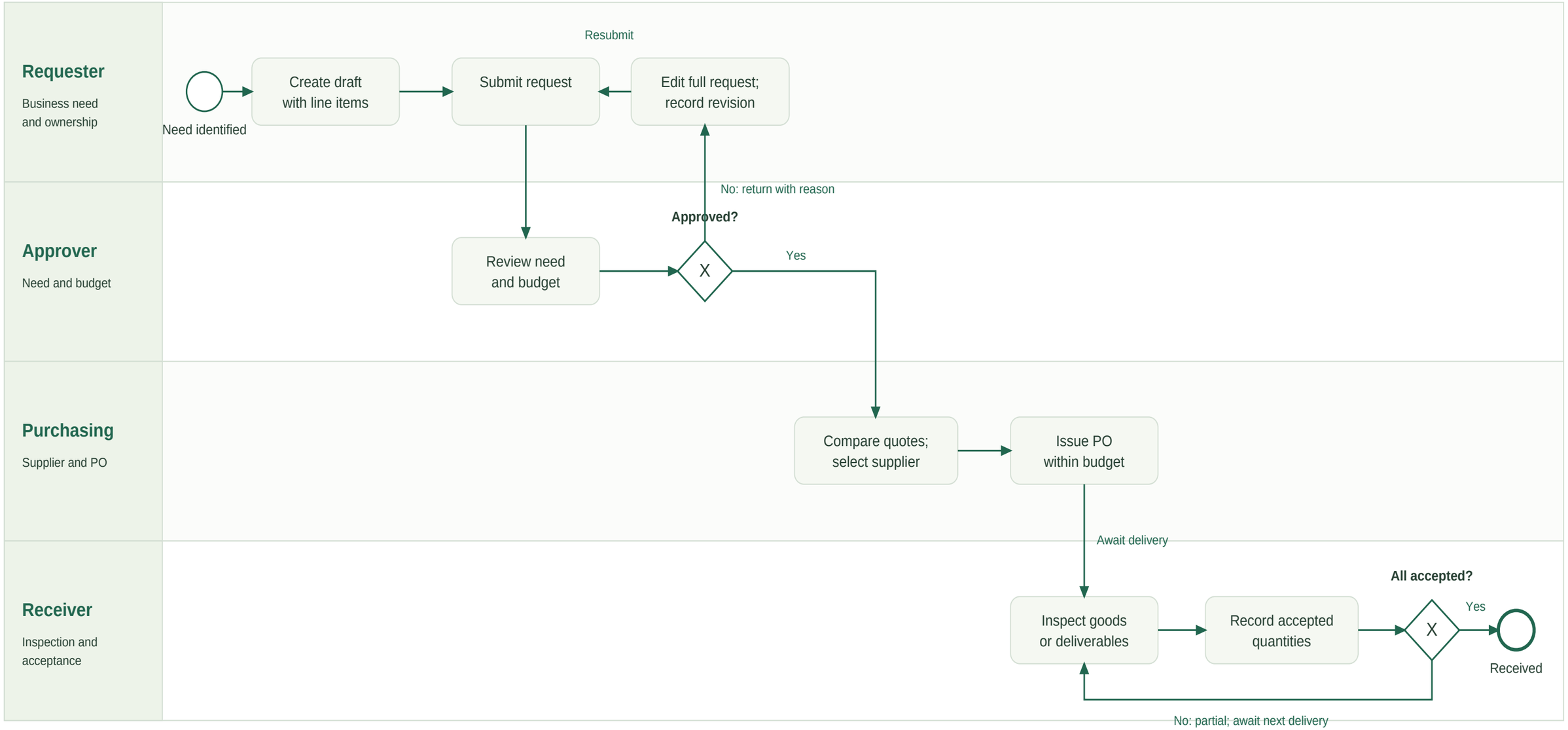
Procurement process

BPMN swimlanes - one internal organization. Supplier contact and physical delivery happen outside the application.



Controls at each handoff

Role and ownership checks | Current state validation | Atomic writes and audit | Unique delivery reference | No over-receipt

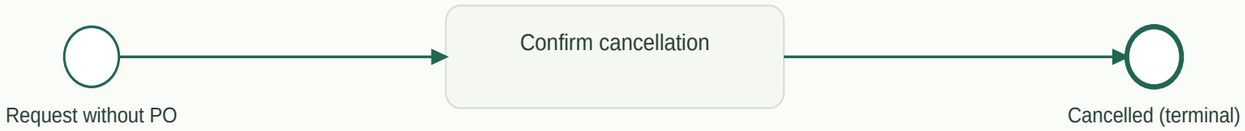
Before a PO exists, owners can edit or cancel. Editing approved content clears approval and starts a new revision.

Exceptions, permissions and scope

Assumptions are explicit so the diagram, WBS and application can be reviewed together.

Cancellation subprocess - Requester lane

The owning requester may cancel a Draft, Submitted, Revision needed or Approved request until a PO is issued.



Business rules

Approval	One stage. No self-approval. Decisions refer to the exact submitted revision; snapshots remain available.
Purchase order	One PO per request. Select a quotation for the approved revision. Copy agreed line prices; stay within budget.
Acceptance	Only inspected, accepted units count. Missing or damaged units stay outstanding until a later delivery.
Duplicates	Exact operation retries return the original result. Changed payloads and stale form versions are rejected.
Revision	Owners edit full drafts, returned or approved requests before ordering. A changed submitted revision clears approval.
Visibility	Requesters see their own requests. The other three roles see all requests, with separate action permissions.

Programming test v4 scope

The revised Procurement brief requires a Python web app with SQLite/PostgreSQL. Chat and mobile are not required.

Included: revision-aware requests, approval, supplier quotation comparison, itemized PO, partial receiving, document input and audit.

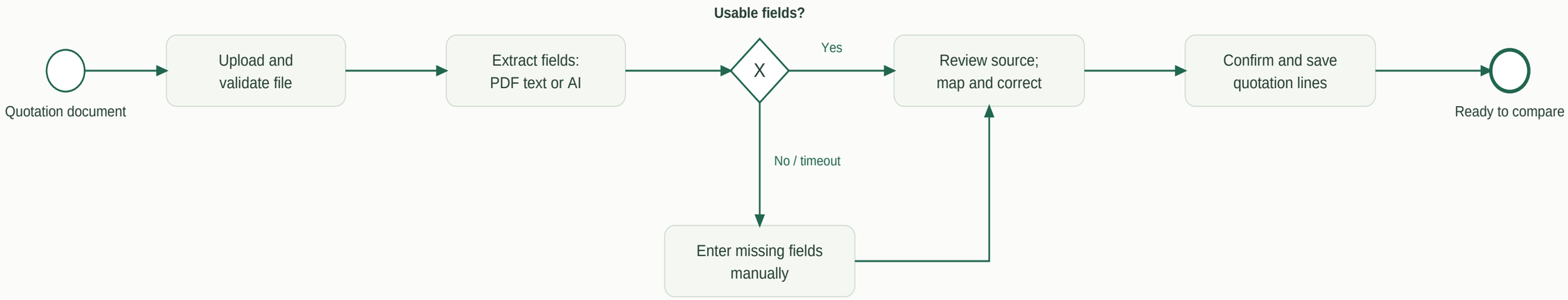
Excluded: warehouse stock, payments, supplier portal and real order dispatch. Optional AI requires configured provider credentials.

Notation: thin circle = start; thick circle = end; rounded rectangle = task; X diamond = exclusive gateway; solid arrow = sequence flow.

Document to quotation form

Purchasing lane - extraction proposes values. Human review and the ordinary approval rules remain authoritative.

Purchasing / current approved revision



File controls

PDF/JPG/PNG; maximum 5 MB; PDF maximum 5 pages; images maximum 12 megapixels. Private storage and authorized download.

AI boundary

Explicit consent for external AI processing. The extractor cannot approve requests, select suppliers or create purchase orders.

Missing or ambiguous values remain unknown. User verifies currency, unit prices, quantities, source evidence and supplier mapping.

Local labeled-PDF parsing is explicitly identified as a parser, not AI/OCR. Manual input remains available when extraction fails.